Hands-On Lab — Generative AI vs. Agentic AI

Workshop: Generative and Agentic AI — Talencia **Duration:** 30 minutes (15 min per lab) **Level:** Beginner | No coding required

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Before You Begin — The Core Idea

Both labs use AI, but they work in fundamentally different ways. Keep this in mind as you work through each one:

	Generative AI	Agentic AI
What it does	Produces content (text, images, code) from a single prompt	Pursues a goal by deciding actions, using tools, and iterating
How it works	One prompt in → one response out	Goal in → AI plans → AI acts → AI observes → AI responds
Knowledge source	Its training data (fixed point in time)	Live tools: web search, databases, APIs
You control	What you ask and how you ask it	The goal; the AI controls the steps
Analogy	A very knowledgeable author writing on demand	A junior analyst who researches and reports back

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Lab 1 — Generative AI (15 minutes)

Tool: ChatGPT — chat.openai.com **Domain:** Healthcare — Patient Communication **Account:** Free account required (sign up with email or Google)

Scenario

You work at a city hospital's communications team. The hospital wants to publish a simple, friendly guide to help patients understand **Type 2 Diabetes** — what it is, how it develops, and what they can do about it. The guide must be readable by someone with no medical background.

Step 1 — Sign In and Start a New Chat

- [ ] Open chat.openai.com in your browser
- [ ] Sign in or create a free account
- [ ] Click **New Chat**

Step 2 — Try the Basic Prompt

Copy and paste the prompt below exactly as written, then press Enter:

Explain Type 2 Diabetes to a patient with no medical background. Keep it under 150 words.

Observe:

- How long did it take to respond?
- Did it use medical jargon or plain language?
- Did it fetch any live information from the internet, or is this from its own knowledge?

Write your observation here:

_____

_____

Step 3 — Improve with a Better Prompt

Now try this more detailed prompt. Notice how adding **context and constraints** changes the output:

You are a friendly healthcare communicator writing for a public hospital in India. Explain Type 2 Diabetes to a patient aged 50+ with no medical background. Use a simple analogy to explain blood sugar. Include 3 practical lifestyle tips. Keep the total response under 200 words and avoid medical jargon.

Observe:

- How is this response different from Step 2?
- Did the analogy help make it clearer?
- The AI never left the chat window to look anything up. All of this came from its training. Is this a strength or a limitation?

Write your observation here:

_____

_____

Step 4 — Try a Variation (Optional, if time allows)

Change the audience and observe how the output shifts:

Rewrite the same explanation but now the audience is a 16-year-old student, not a patient. Keep it casual and engaging.

Lab 1 Reflection

Answer these questions before moving to Lab 2:

1. Did ChatGPT ever search the web or visit any website to answer you?

[ ] Yes [ ] No

2. Could you verify where this information came from?

[ ] Yes, it cited sources [ ] No, it gave no sources

3. In one sentence, what does Generative AI do?
4. _____

Lab 2 — Agentic AI (15 minutes)

Tool: Perplexity AI — perplexity.ai **Domain:** Business & Finance — Market Research **Account:** No account required (works in browser directly)

Scenario

You are a junior business analyst at a consulting firm. Your manager has asked you to prepare a **quick briefing on the current state of the electric vehicle (EV) market** before a client meeting tomorrow. You need recent data, key players, and trends — not textbook knowledge from two years ago.

Step 5 — Open Perplexity AI

- [ ] Open perplexity.ai in your browser (no login needed)
- [ ] You will see a search/chat bar at the centre of the page
- [ ] Look at the top of the page — notice it says **"Focus"** options (Web, Academic, etc.). Leave it on the default.

Step 6 — Ask Your First Research Question

Type the prompt below and press Enter:

What is the current state of the global electric vehicle market in 2025? Include key players, recent growth figures, and major trends.

Watch carefully as it responds:

- [ ] Do you see it show "Searching…" or a list of sources appearing on the side?
- [ ] Are there numbered citations [1], [2], [3] in the response?
- [ ] Can you click on those citations and see the actual source websites?

Write what you notice:

_____

_____

Key insight for the facilitator to highlight: Perplexity did not answer from memory. It autonomously decided what to search, retrieved live pages from the web, read them, and synthesised the answer. That is agentic behaviour — it took actions (web searches) to complete your goal.

Step 7 — Ask a Follow-Up That Forces Multi-Step Reasoning

Now ask a question that requires the AI to connect multiple pieces of information:

Based on current EV market trends, which 3 countries should a new EV battery startup target first, and why? Consider market size, government incentives, and infrastructure readiness.

Observe:

- The AI now has to reason across multiple factors, not just retrieve one fact.

- Did it give a structured recommendation with reasoning?
- Are the recommendations backed by cited sources?

Write your observation here:

_____

_____

Step 8 — Test Its Live Knowledge

Ask something that a model trained two years ago could not answer accurately:

What are the top 3 most significant EV news stories from the last 3 months?

Observe:

- Are the stories recent (check the dates in the citations)?
- ChatGPT (from Lab 1) would struggle with this question. Why?

Write your observation here:

_____

_____

Lab 2 Reflection

1. Did Perplexity search the web to answer your questions?

[ ] Yes [ ] No

2. Could you verify where the information came from?

[ ] Yes, it cited sources [ ] No, it gave no sources

3. In one sentence, what does Agentic AI do differently?
4. _____

Comparison — What Did You Experience?

Fill in this table based on your observations from both labs:

Question	Lab 1 — ChatGPT (Generative)	Lab 2 — Perplexity (Agentic)
Did it search the internet?		
Did it show sources / citations?		
Could it answer questions about events from last week?		
Did you control every step, or did it take actions itself?		
Was the response instant or did it "work" before answering?		

Key Takeaways

- **Generative AI** is like a brilliant author: it produces high-quality content on demand, but everything it knows was baked in during training. It is a single step: prompt in, content out.
- **Agentic AI** is like a junior analyst with internet access: it receives a goal, decides what actions to take (search, retrieve, compare), executes those actions, and then reports back. It is a loop: plan → act → observe → respond.
- Neither is better overall — they solve different problems:
  - Use **Generative AI** when you need to create, draft, summarise, or rewrite content.
  - Use **Agentic AI** when you need to research, investigate, or make decisions with real-world, up-to-date data.
- In real-world AI systems, both are often combined: an agent uses tools to gather information, then a generative model turns that information into a final response.

Bonus Challenge (If You Finish Early)

Try asking ChatGPT the same market research question you gave Perplexity in Step 6:

What is the current state of the global electric vehicle market in 2025? Include key players, recent growth figures, and major trends.

Compare the two answers side by side:

- Which is more current?
- Which has verifiable sources?
- Which would you trust more for a real business briefing, and why?

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